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The Microbiome in Gastrointestinal Cancers

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INTRODUCTION

Not long after the introduction of the germ theory of disease by Louis Pasteur and Robert Koch, Izmar Isidor Boas and Bruno Oppler began to describe bacteria within the luminal contents of the stomach of patients with gastric cancer and specific species that were notably absent from the contents of healthy controls. This noted “colonization” of the stomach with bacteria was thought to potentially relate to physiologic changes within the stomach itself associated with carcinogenesis.¹ Given the inherent limitations of the time, however, these associations could only be made through microscopic evaluation of stomach or stool contents or by culturing these contents within various media—leading to notable biases in observations and clouding more in-depth study. A century later, we are continuing to discover and describe the role of bacteria in cancer initiation, progression, and evasion of contemporary therapeutics (cytotoxic chemotherapy, immunotherapy, and radiation therapy). In the last 3 decades, with the introduction and popularization of next-generation sequencing (NGS) technologies, an omics-based approach to the microbiome has led to a vastly more comprehensive and detailed descriptions of human microbiology and downstream insights into their role in gastrointestinal (GI) malignancies outlined here.

Although several hypotheses have been suggested across disease histologies, the role of bacterial or viral species in the development of GI malignancies continues to be elucidated. To date, several viruses and bacteria including *Helicobacter pylori*, hepatitis B and C, HIV human papillomavirus, Epstein–Barr virus, human herpesvirus type 8, human T-cell lymphotropic virus type 1, *Opisthorchis viverrini*, *Clonorchis sinensis*, and *Schistosoma haematobium* have been clearly linked to carcinogenesis.² These microbes, and others under active study, are thought to affect cancer outcomes through alterations of metabolomic pathways, induced changes in the host immune system, and alterations

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in the pharmacokinetics of anticancer agents. Beyond the epidemiologic or culture-based systems of study, the contemporary hypotheses of the role of microbes in these systems are heavily dependent on accurate descriptions of the host immune system and the tumor microenvironment.

The introduction of NGS techniques was a disruptor in microbial research. Using NGS techniques, it is now possible to definitely and precisely describe the microbial compositions of gut luminal contents, tumors, healthy tissues, and circulating plasma. One of the earliest of these techniques sequences the RNA from the 16s ribosomal component of the 30s ribosomal subunit of prokaryotic species. These RNA can be sequenced using NGS technology to provide the rapid and accurate identification of bacterial species. In addition, several software packages have recently been developed to identify bacterial, viral, or fungal sequence reads in transcriptomic, exomic, and genomic sequencing of human tissue. Classically discarded as a quality control step in standard human sequencing, these tools leverage discarded raw sequence reads aligned to microbial sequences or various bacterial taxa or viruses.

Importantly, these observations describe a point in time of a complex system and are independent of culture growth kinetics or other microbial factors that were inherent biases in years passed. Simultaneously, recognition of the importance of the immune system in cancer surveillance and suppression has been increasingly appreciated in the last decade. This has led to the description of a critical axis that exists between the host microbiome, immune system, and cancer-specific outcomes. Ultimately, as the importance of these bacterial and viral species throughout the body is understood, therapeutics leveraging these observations to improve care is being developed, providing the opportunity for these advances in knowledge to improve patient outcomes.

Here, we review work demonstrating a variety of associations between GI malignancies (gastric, pancreatic, and colorectal adenocarcinoma [CRC]) and the gut and tumoral microbiome. We also review the role these microbes play in evasion of therapeutics targeting these cancers. Last, hypotheses and early data supporting the mechanistic underpinnings of these associations are described. Understanding of these factors is critical to leveraging work associating microbial changes with disease to tailor dietary or novel targeted therapies to improve health across the spectrum of malignant disease.

THE MICROBIOME AND CARCINOGENESIS

The host microbiome has long been considered a potential contributing factor to the development of a variety of malignancies. To date, several bacteria and viruses have been clearly delineated as contributing to carcinogenesis. These contributions occur through direct cytotoxic effects disrupting autophagy and apoptosis pathways as well as modulating oncogenic signaling pathways. Observed effects vary by bacterial taxa and cancer histology of interest. Moreover, beyond individual taxa associations, broader microbial community level shifts have the potential to modulate cancer risk via microbial metabolites and the overall health of the gut-immune axis.

Colon Polyps and the Microbiome

A CRC is classically thought of as the progression of normal colonic epithelium to polyp formation through progressive cellular changes that eventually lead to an adenoma and continued cellular alterations resulting in the CRC. Microbes have been shown to correlate both with CRC and their precursor polyps. At the polyp stage, these tissues have been noted to have higher rates of proteobacteria as well as lower rates of bacteroidetes when compared with healthy colonic tissue.³ In studying the gut microbiome of patients with colon polyps, a general bacterial dysbiosis as well as fungal signatures of the ascomycota/basidiomycota ratio was noted along with the increased proportions of opportunistic fungi *Trichosporon* and *Malassezia*.^{4,5} Although these correlations are intriguing, a causal role of these gut microbiomes in CRC development has, to date, only been suggested in murine models via the administration of stool from patients with CRC leading to higher rates of high-grade dysplasia and polyp formation.⁶ Furthermore, the impact of the gut microbiome on the host immune system was reflected in upregulation of inflammatory pathways and intestinal recruitment of T-helper-1 (Th1) and Th17 cells. Lastly, in animal models with induced colorectal carcinogenesis, germ-free rats grew fewer and smaller tumors when compared with rats with conventional gut microbiota suggesting a critical role of the microbiome in formation of colonic polyps.⁶

Colorectal Adenocarcinoma and the Gut Microbiome

CRC is the third most commonly diagnosed malignancy worldwide and the third most common cause of cancer-related mortality. It has been associated with diets high in red meat, low-fiber diets, alcohol consumption, smoking, and obesity. Importantly, CRC has been rising in incidence (especially in younger populations) and hypothesized to have a potential link to environmental exposures given this recent and sudden rise. Moreover, several studies are underway to correlate this increase in incidence with the gut microbiome given its strong link to environmental exposure and potentially with CRC development.

The flora of the gut under normal physiologic conditions varies from proximal to distal and is reflective of one's exposome—their location, diet, medications, and other lifestyle factors.⁷ These changes are reflected in enzymatic activity, pH, and fermentation of luminal contents. The resulting changes are reflected in various genera of bacteria being abundant within the GI tract and colon. Beyond red meat, epidemiologic associations have been made with high-fat diets which have been shown to increase sulfate-reducing bacteria. These bacteria are critical to the transformation of primary to secondary bile acids shown to be potentially related to carcinogenesis.⁸ Conversely, butyric acid and short-chain fatty acids (generated from fermented fiber) have been shown to be protective of CRC development.⁹ The role of metabolites has also been shown to affect the function of p53, converting it from a tumor suppressor to oncogene depending on the levels of luminal gallic acid.¹⁰ These well-described associations between CRC and diet have therefore led to intensive study of the role of the gut microbiome in the development of CRC.¹¹ Various reports have suggested the most abundant genera of bacteria to be *Bacteroides*, *Prevotella*, and *Ruminococcus* in CRC patients.¹² Importantly, markers of a “healthy” microbiome remain elusive, although markers of carcinogenesis and dysregulation are increasingly well described.

As noted in other malignancies, the gut microbiome and metabolome have been shown to correlate with colorectal cancer in a stage-specific manner.^{13,14} In the case of CRC, *Fusobacterium nucleatum*, and *Solobacterium moorei* were present in higher abundance in patients with later-stage CRC. *Bacteroides fragilis* and colibactin-producing *Escherichia coli* have also been associated with CRC and have been suggested to play a potential causal role in initiation and progression of these tumors.^{15–18}

It has been demonstrated that *B fragilis* enterotoxin plays a multifactorial role in oncogenesis. This enterotoxin's mechanism is similar to colibactin produced by *E coli*^{19,20} and has the ability to induce direct DNA damage via reactive oxygen species. This then leads to the Th17 recruitment which produces IL-17 and downstream NF- κ B (Nuclear factor kappa-light-chain-enhancer of activated B cells) signaling as well as E-cadherin, β -catenin, and STAT3 (Signal transducer and activator of transcription 3).^{21,22} This induces a pro-inflammatory setting stimulating IL-8, TGF β (Transforming growth factor beta), C5a (complement component 5a), leukotriene B4, and growth-related oncogene- α , featuring immature myeloid cells that lead to an oncogenic microenvironment favoring cancer initiation and progression.^{15,23,24} At the same time, *B fragilis* enterotoxin plays a role breaking down mucus overlying the colonic epithelial cells allowing for the adhesion of *B fragilis* and other opportunistic species.²⁵

At a more global level, the administration of stool from patients with CRC was able to induce higher rates of dysplasia and polyp formation suggesting a causal role of the gut microbiome in colorectal cancer initiation.⁶ Beyond direct invasion, toxin production, and DNA damage, the effect of the immune system is significant with the host immune system recognizing various microbial markers. Recognition of these microbe-associated molecular patterns (MAMPs) leads to the activation of NOD (nucleotide-binding oligomerization domain)-like receptors and Toll-like receptors (TLRs) that lead to the regulation of inflammatory pathways and the proliferation of various immune compartments.^{26–28} In particular, TLR2 has been suggested to play a role suppressing local immune response and leading to protection of the colonic epithelium.^{29–31} The importance of the interface between host systemic immunity and the gut microbiome in these observed effects was demonstrated in animal models of CRC carcinogenesis, germ-free rats grew fewer and smaller tumors than those with a common gut microbiota.³²

Gastric Adenocarcinoma and the Gut Microbiome

Gastric cancer is the fourth most common cancer worldwide and the second most common cause of cancer-related death. Chronic *H pylori* infection has been associated with the development of peptic ulcer disease and gastric adenocarcinoma.

Gastric adenocarcinoma development has been correlated with both local gastric luminal contents³³ and the colonic gut microbiome.³⁴ In the case of luminal contents, the most robustly described is the case of *H pylori* that has been clearly and mechanistically associated with the development of peptic ulcer disease. Intriguingly, the association between *H pylori* infection and gastric cancer risk was shown to be strongly associated with patient ancestry in a cohort of Colombian patients.³⁵ Although those with Amerindian ancestry were at a high risk of gastric cancer in the setting of *H pylori* infection, it was

relatively benign in a cohort of those with African ancestry suggesting germline risk is strongly intertwined with this risk profile.

The demonstrated mechanism behind the development of gastric adenocarcinoma secondary to *H pylori* infection is the injection of cytotoxin-associated gene A (CagA) and vacuolating toxin A (VacA) into gastric epithelial cells. In the case of CagA, this leads to the loss of cell polarity, induction of inflammation (interferon- γ , TNF [Tumor necrosis factor]- α , IL-1, IL-1 β , IL-6, IL-7, IL-8, IL-10, and IL-18), disruption of epithelial junctions, and oncogenic signaling initiation (ERK/MAPK [extracellular signal-regulated kinases/mitogen-activated protein kinase], PI3K/Akt [Phosphoinositide 3-kinase/Protein kinase], NF- κ B, Wnt [Wingless-related]/ β -catenin, Ras [Rat sarcoma virus], sonic hedgehog, and STAT3) that induce an oncologic inflammatory milieu.^{36–38} In addition, VacA induces vacuolation and autophagy of epithelial cells, increased MAP kinase activity, VEGF (Vascular endothelial growth factor), and Wnt/ β -catenin.^{39–43}

Aside from the specific associations and mechanism behind *H pylori* carcinogenesis, more global dysbiosis has been noted in the gastric luminal contents of these patients. These changes have been noted to change across stages of development and progression of gastric adenocarcinoma. This work has identified overall compositional changes in gastric luminal contents as well as enrichments of *P stomatis* (Peptostreptococcus stomatis), *Dialister pneumosintes*, *S exigua* (Slackia exigua), *Parvimonas micra*, and *Streptococcus anginosus*.³³

Beyond these local associations of gastric luminal contents and local effect, the role of the gut microbiome in gastric cancer carcinogenesis has been elucidated recently. Specifically, *Clostridium* and *Fusobacterium* species have been noted in higher abundance in patients with gastric cancer as compared with the general population.⁴⁴ These findings were further validated along with a more comprehensive report of differential abundance in patients with gastric cancer described by Zhang and colleagues.³⁴ In the representative of the translational potential of this work, both *Lactobacillus* and *Megasphaera* were found to be potentially useful as markers for gastric adenocarcinoma detection. These associations must be taken in light of the significant work in other disease histologies and associations between colonic gut flora and disease initiation and progression. Further work will ultimately be needed to define whether these associations are correlative or causal.

Pancreatic Adenocarcinoma and the Gut Microbiome

Pancreatic cancer is the 12th most common cancer worldwide and the 7th cause of cancer-related death. It has an almost inherently poor prognosis with a median survival of 9 months. It is associated with smoking, obesity, alcohol, chronic pancreatitis, and type 2 diabetes.

Epidemiologic data have intriguingly linked periodontal disease with pancreatic ductal adenocarcinoma (PDAC) incidence across several cohort studies with a wide range of reported risks.^{45–49} Given these associations, directed analyses of the oral microbiomes of patients with pancreatic cancer and healthy controls demonstrated alterations in a variety of bacterial species including *Neisseria elongate*, *Streptococcus mitis*, *Fusobacterium*, *Leptotrichia*, Firmicutes, and *Porphyromonas*. These were shown to be differentially

abundant between groups of cases and controls.^{50–54} Intriguingly, one of these studies also noted circulating antibodies to these oral bacteria correlated with risk of PDAC.⁵⁴

Similar to previously described associations in colorectal and gastric cancer, it has been noted that there are differential abundances of bacteria in fecal samples of patients with PDAC as compared with healthy controls. This is perhaps not surprising given the critical role the colonic gut microbiome plays in normal pancreas physiology.⁵⁵ To date, however, this has only been completed in a single study demonstrating an increase in proteobacteria in PDAC and a depletion in short-chain fatty acid synthesis association modules, with an increase in modules associated with bacterial virulence.⁵⁶ Ultimately, validation and further study is needed to solidify these observations and study any potential causal role of the gut microbiome in PDAC.

Although minimal data exist for associations between the colonic microbiome and PDAC, there are correlative trials associating gastric *H pylori* infection with PDAC—although these are significantly confounded by lifestyle factors such as smoking and obesity.⁵⁷ The proposed mechanism lies behind the upregulation of *KRAS*, mutated in the majority of PDAC, by *H pylori*.⁵⁸ In addition, *H pylori* has the potential to upregulate Bcl-xL (B-cell lymphoma-extra large), MCL-1 (Induced myeloid leukemia cell differentiation protein), surviving, c-myc (Myelocytomatosis), and cyclin D1 which all have the potential to contribute to PDAC carcinogenesis.^{59–61}

As described in CRC, the role of MAMPs in PDAC has been preliminarily studied as well. The induction of the immune system via MAMPs has been shown to lead to pancreatitis and potentially progress to pancreatic cancer.^{62,63} Although *F nucleatum*, as in CRC, is also a negative prognostic factor in PDAC.⁶⁴ Finally, the taste receptor 2 member 38 (T2R38) has been shown to be expressed in PDAC. Moreover, *Pseudomonas aeruginosa* is a T2R38 ligand that has been shown to lead to invasion and metastasis via ABCB1.⁶⁵

THE MICROBIOME IN DISEASE PROGRESSION

Beyond cancer initiation and carcinogenesis, bacteria have the potential to induce metastatic spread. These effects are possible both in the disruption of vascular barriers that may initially confine malignancies and alterations in the immune system both globally and locally that diminish its routine immune surveillance functions that act to suppress metastatic spread. For clinicians caring for solid tumors such as the GI malignancies reviewed here, this is a particularly critical step in disease progression as this classically represents the point at which patients move from being curable to incurable. As such, the potential future role of the microbiome in prognosis, surveillance, and therapy is immense.

Colorectal Adenocarcinoma

Beyond gut microbial associations, the specific bacterial taxa previously mentioned (*F nucleatum*) is oftentimes found within colorectal cancers themselves. These observations suggest a potential role in carcinogenesis for a subset of patients with colorectal cancer.^{66–71} Beyond associations with colorectal cancer, a specific phenotype seems to be associated with *F nucleatum* positive CRC. These tumors are predominantly right sided

and associated with an overall poor prognosis.^{72–75} In the case of *F nucleatum* positive rectal cancer, clearance of tumoral *F nucleatum* is associated with improved recurrence free survival as compared with those that remain positive post-neoadjuvant chemoradiotherapy, demonstrating tumoral levels to be a modifiable risk factor that has the potential to improve cancer-specific outcomes. Although numerous groups have posited hypotheses as to the mechanisms behind these observations, the role of *F nucleatum* in carcinogenesis is likely multifactorial.

Of these hypotheses, the role of Fap2 (fusobacterial Gal-GalNAc-binding lectin), a virulent epithelial adhesin, is often suggested as a potential mechanism to explain these observations. Fap2 permits adhesion to a CRC polysaccharide and generates a FadA (Fusobacterium Adhesions Gene A) adhesion complex capable of stimulating Wnt/ β -catenin, leading to the increased expression of oncogenic and inflammatory responses.^{76–78} Treatment of murine models of CRC with *F nucleatum* was able to mimic similar transcriptional and inflammatory pathways seen clinically.^{68,79} Furthermore, incubation of CRC cell lines with *F nucleatum* before injection into mice led to increased growth rates of incubated cell lines as compared with control.^{76,80}

As detailed above, several bacterial taxa have been implicated in the development of CRC. Beyond these associations, overlapping and unique species also play a role in the progression of disease. In the case of *F nucleatum*, murine models of colorectal cancer liver metastasis have been shown to show a less virulent phenotype with the eradication of *F nucleatum* from the murine gut.⁸¹ Interestingly, these tumors that are *F nucleatum* positive have *F nucleatum* present in positive lymph nodes as well as liver metastases, further strengthening the potential association between its presence and disease progression and metastasis.⁸¹

Similarly, the development of metastasis via disruption of the vascular barrier and formation of a pre-metastatic niche has been shown to be driven, in murine models by gut bacteria (potentially *E coli* C17). Moreover, circulating markers of gut vascular permeability (plasmalemma vesicle-associated protein-1) were shown to correlate with outcomes in clinical specimens and again be diminished with eradication of gut bacteria.^{82–86}

Pancreatic Adenocarcinoma

Apart from limited data describing the role of the gut microbiome in pancreatic cancer, there has been a significant body of work recently describing the role of the tumoral microbiome's role in the microenvironment of PDAC. In the case of PDAC, the tumoral microbiome seems to play two distinct roles—in the modulation of immune response as measured by T-cell activation^{87–89} and in the inactivation of common cytotoxic agents such as gemcitabine.⁹⁰ Importantly, these observations are tied to outcomes of patients with PDAC, as long-term survivorship was associated with a more diverse tumor microbiome, specific bacterial taxa (*Saccharopolyspora*, *Pseudoxanthomonas*, and *Streptomyces*), and a more robust microbially driven immune response in the tumor microenvironment.

THE MICROBIOME AND RESPONSE TO THERAPY

Intriguingly, a variety of bacterial species have been shown to alter response to various therapeutics. In the case of immune checkpoint blockade (ICB), this is somewhat expected given the known significant impact the gut microbiome has on modulating the host immune system. A similar effect has been seen locally as well, with the tumor microbiome modulating response to immunotherapies. Lastly, the degradation of cytotoxic therapies by tumoral bacteria have been shown to decrease the efficacy of cytotoxic chemotherapy. These initial findings lay the groundwork for the potential of microbial modulation to augment other therapies—as has been seen in the case of fecal transplantation inducing response to ICB in melanoma.⁹¹

Colorectal Adenocarcinoma

A subset of colorectal cancers, microsatellite mismatch repair deficient, are known to be responsive to the ICB.⁹² Response to ICB, however, is not uniform. Similar to other histologies, this heterogeneity has been hypothesized to, at least partially, correlate with specific gut microbial communities and changes.⁹³ Studies quantifying this association and overlap with gut signatures predictive of response in other histologies, however, are ongoing in the setting of promising preclinical models.⁹⁴

Although microbial markers of response to checkpoint blockade are actively being studied, markers of response to cytotoxic therapies are increasingly well described. In the case of *F nucleatum*, previously associated with carcinogenesis, it has also been associated with chemoresistance in CRC.^{14,74} This induced chemoresistance has been noted with both oxaliplatin and 5-FU via the activation of autophagy pathways and innate immune signaling via TLR4 and MyD88.⁷⁴ These effects were shown to be abrogated in preclinical models via the modulation of the gut microbiome with antibiotic therapy, again suggesting a causal rather than correlative association.⁹⁵

Pancreatic Adenocarcinoma

As previously noted, bacteria within PDAC tumors have the ability to inactivate cytotoxic agents such as gemcitabine. These initial analyses importantly provide insights in prognostication while also suggesting potential future novel treatment strategies. Moreover, unique neoantigen properties of long-term survivors noted by Balachandran and colleagues demonstrate circulating levels of MUC16 (CA125) and T-cell reactivity are associated with survival and that their loss is associated with relapse.⁸⁷ Pushalkar and colleagues were able to demonstrate the tumoral microbiome acts to suppress monocytic cellular differentiation that leads to T-cell anergy.⁸⁸ These mechanistic associations are important keys as the correlative versus causal role of the microbiome continues to be delineated while also allowing for more directed hypothesis generation in other disease sites.

SUMMARY

As outlined above, the interplay between the gut and tumoral microbiome, host immune responses, carcinogenesis, and evasion of therapeutics is a complex system with significant cross-talk between these interrelated variables. Although these associations begin to be

deconvoluted, the perennial question in microbiome research exists between causation and correlation. Hypotheses exist for several specific bacteria, and we see overlap between taxa or pathway associations with various disease histologies—short-chain fatty acids, *H pylori*, and *F nucleatum* are clear examples of this. Moreover, with the initiation of clinical trials modulating the microbiome through fecal transplantation, prebiotics, or other novel methods, of these associated mechanisms will be tested with continued rigor and continue to provide further insights.

Microbial modulation via phase I studies of fecal transplantation in melanoma patients demonstrated the potential efficacy of gut microbial alteration in optimizing responses to therapy.^{91,96} This work provided clear insight that fecal transplantation was able to increase tumor immune infiltrates and induce response in 30% to 40% of otherwise checkpoint blockade refractory patients. Although these findings were noted in cutaneous malignancies, these therapies have the potential to improve outcomes in checkpoint responsive GI malignancies and warrants study in cytotoxic therapies. In the case of fecal transplant technologies, inherent difficulties in scalability are likely insurmountable on a population level. These studies, however, importantly inform the development of directed microbial modulation as well as dietary and prebiotic modulation of the gut microbiome.

An increasing interest in the use of changes in diet and/or the use prebiotics (non-digestible agents used to promote beneficial microbes) has driven directed work studying the potential benefits of these agents in improving response to various therapies. Dietary associations with various malignancies have long been described, such as associations between red meat intake and colorectal cancer.¹¹ In this case, however, the use of nutritional supplements such as fiber⁹⁷ or soluble vitamins such as magnesium⁹⁸ has been shown to improve response with these dietary effects seen during their treatment window. This allows clinicians to make a more targeted and directed intervention beyond global dietary changes. Intriguingly, probiotics (ingested live bacteria) decrease gut microbial diversity and have been associated with worse outcomes in immunotherapy-treated patients.⁹⁷

Perhaps the most immediate potential clinical application of the microbiome lies in the ability of gut, tumoral, or circulating microbes to act as screening or prognostic tests at population or individual levels. Recent analysis of the cancer genome atlas has shown histology and stage-specific circulating microbial signatures detectable across disease sites.⁹⁹ These microbes exist in tissue and/or blood oftentimes already collected as part of standard of care and could be queried through scalable targeted assays or can be sequenced as part of exploratory research protocols. Furthermore, with continued optimization of gut microbial collection techniques, patients can now submit stool samples remotely for 16s or whole-genome sequencing. The ability of these or other markers to augment current standard of care tests, however, remains unknown and will require independent study for each disease histology.

As the microbiome is increasingly recognized as critical to human health and normal immune function, its importance to malignant disease development, progression, and response to therapy is increasingly clear. Moving forward, clinical trials and careful prospective studies are needed to begin to integrate these findings into clinical practice.

As the early adoption of various treatment modalities is initiated, it will be increasingly important to actively study these patients to further improve and learn from these early studies. Ultimately, the last decade has shown continued interest and momentum behind the study and utilization of microbial modulation to improve cancer care.

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KEY POINTS

- The gut microbiome and host immune system are in a dynamic homeostasis whose alterations have been associated with the development and progression of a variety of malignancies.
- The tumoral microbiome plays a role in the tumor microenvironment modulating immune responses to tumors and has the potential to affect the pharmacodynamics of some cytotoxic therapies.
- Further work is needed to delineate mechanisms behind these observations and translate them to clinical benefit for patients.

CLINICS CARE POINTS

- The gut microbiome is an emerging factor in the development and progression of gastrointestinal malignancies.
- While emerging therapeutics in benign disease such as refractory clostridium difficile infection are approaching approval in the United States, therapeutic and diagnostic strategies are still under development.
- Commercially available fecal profiling offer little insight into one's overall microbial health and their results should be interpreted with caution.